

Session 6: Data visualisation and ggplot

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- 2 Data visualisation
- 3 ggplot()
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- 5 Working on projects

Introduction

Learning goals of today

- Short repetition from last week

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- Crash course in data visualisation

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- Short repetition from last week
- Crash course in data visualisation
- Intro to ggplot

Repetition: Useful commands

- `group_by()`: Grouping by variable (for manipulation)

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Repetition: Useful commands

- `group_by()`: Grouping by variable (for manipulation)
- `join()`: Join two data sets on identifier
- `arrange()`: Sort all data by variable(s)
- `pivot()`: Re-order data between rows and columns

Data visualisation

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- But all the data in the world is meaningless if it is not understood
- Humans are visual creatures
- Data visualisation is the fastest way for humans to digest complex data

Famous people in data viz: Florence Nightingale

- Florence Nightingale

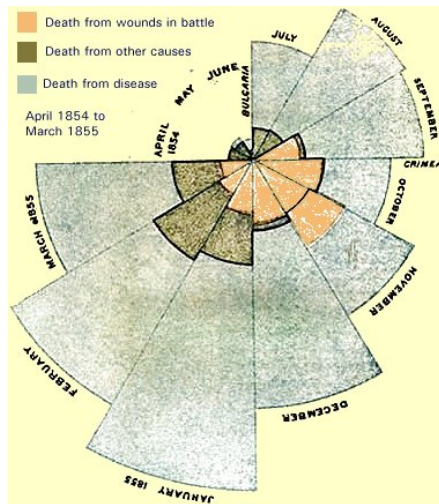
Famous people in data viz: Florence Nightingale

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- Manager and nurse educator in the British Army

Famous people in data viz: Florence Nightingale

- Florence Nightingale
- Manager and nurse educator in the British Army
- Became famous for data visualisation of Crimean War

British Casualties in 1854



Famous people in data viz: Edward Tufte

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- American Statistician
- The Visual Display of Quantitative Information
- Famous for data visualisation principles

Some of Tufte's principles for good data vis

- Show the data

Some of Tufte's principles for good data vis

- Show the data
- Maximize data-ink ratio

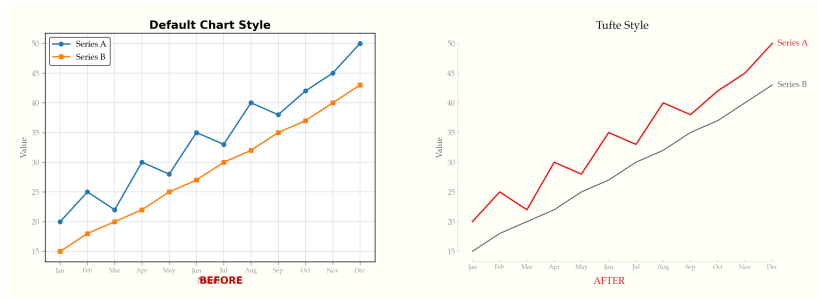
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- Show the data
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- Minimizing chartjunk

Some of Tufte's principles for good data vis

- Show the data
- Maximize data-ink ratio
- Minimizing chartjunk
- Graphical integrity

An example of Tufte's principles



Source: https://raw.githubusercontent.com/caylent/tufte-data-viz/main/_docs/before-after.png

Some further reading

- Data visualization in R

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- R Graph Gallery

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Some further reading

- Data visualization in R
- R Graph Gallery
- Modern Data Visualization with R
- Data Visualization in R (Geeks for Geeks)

ggplot()

Introduction to ggplot()

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- load using library("ggplot2")

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Basic components of ggplot() function

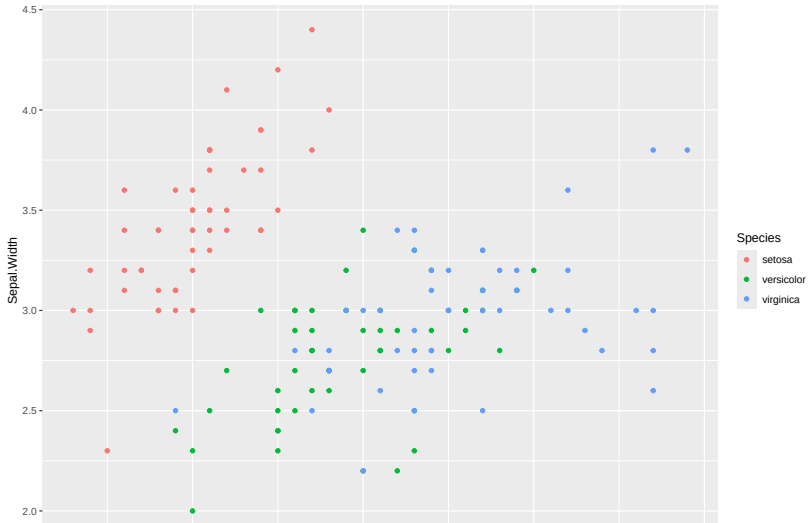
- call function: `ggplot(...`
- add data: `data = ...`
- add aesthetics `aes(x = ... , y = ... , color = ...)`
- add layer that you want to visualise, like points + `geom_point()`

Simple example of ggplot() - Code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set
ggplot(data = iris,
       aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```


Simple example of ggplot() - Graph



Task: Simple scatter plot with your data

- Try to create two simple scatter plot in your data using ggplot()

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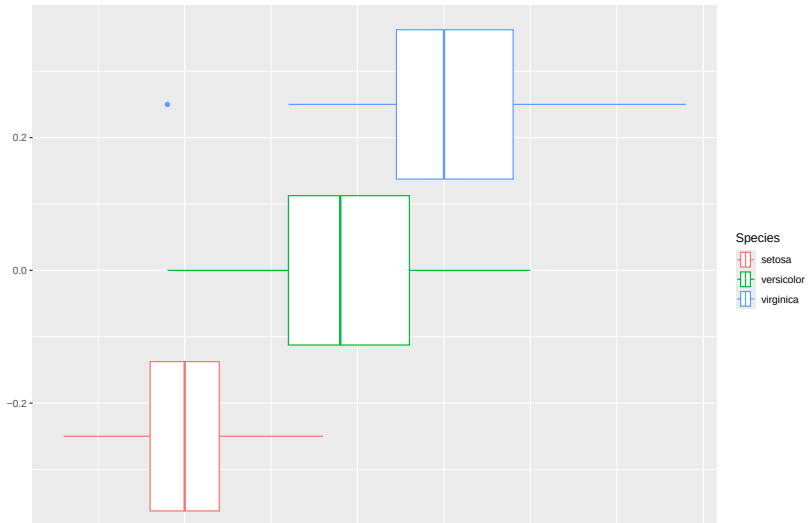


Simple example of ggplot() - Alternative code

```
# Load ggplot2
library(ggplot2)

# Boxplot using iris data set
ggplot(data = iris,
       aes(x = Sepal.Length,
           color = Species)) +
  geom_boxplot()
```

Simple example of ggplot() - Alternative graph



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- I changed the plot entirely by using a different layer
- Different layers are useful for different purposes

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